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CASE REPORT

Toxicology; Pathology/Biology

Homicidal paraquat poisoning: Poisoned while drinking

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Abstract

The incidence of paraquat poisoning has significantly decreased with the addition of odorizer and emetics to the liquid concentrate. Paraquat poisonings are usually attributed to suicidal and accidental or occupational exposure. Here, we report an unusual fatal case of homicidal paraquat poisoning. An intoxicated, a 37-year-old man consumed a mixture of white wine and paraquat prepared by his wife. This resulted in intermittent vomiting, which he attributed to being intoxicated. The man was admitted to the hospital for treatment 3 days later. Due to the lack of knowledge of paraquat exposure, the man did not receive effective treatment and died of respiratory failure 22 days later. High-performance liquid chromatography-tandem mass spectrometry (HPLC-MS/MS) was applied to detect paraquat in 16 postmortem specimens: kidney (1.31 ug/g), urine (0.91 ug/ml), liver (0.62 ug/g), lung (0.39 ug/g), muscle (0.35 ug/g), bile (0.32 ug/ml), heart (0.28 ug/g), brain (0.22 ug/g), pancreas (0.22 ug/g), spleen (0.18 ug/g), cardiac blood (0.15 ug/ml), cerebrospinal fluid (0.14 ug/ml), pericardial effusion (0.12 ug/ml), pleural effusion (0.09 ug/ml), peripheral blood (0.08 ug/ml), and vitreous humor (0.06 ug/ml). The highest concentration of paraquat was detected in the kidney followed by the urine in all tissues and body fluids. At present, although the cases of paraquat poisoning have decreased, the high mortality rate resulting from its irreversible lung damage and respiratory failure makes paraquat poisoning, especially occult paraquat poisoning, still needs to be carefully identified in forensic practice and clinical diagnosis.

KEYWORDS
forensic pathology, forensic toxicology, paraquat, paraquat distribution, poisoning

Highlights

- We report a rare fatal case of homicidal PQ poisoning by oral administration.
- We present the PQ concentrations of 16 postmortem specimens to accumulate data on PQ distribution.
- We also reviewed the previous studies on detection of paraquat in postmortem specimens.
- This case emphasizes the problem of occult paraquat poisoning in forensic practice.

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1 | INTRODUCTION

Paraquat (1,1'-dimethyl-4,4'-bipyridylium dichloride; PQ) is an extraordinarily effective, fast-acting, and nonselective contact herbicide, which can kill a diversity of grass and dicot weeds [1]. Since its introduction in the market in the 1960s, PQ has been widely used in many countries [2]. But because of extremely high mortality, the use of PQ has been restricted in many countries [1]. Pesticide poisoning accounts for about one-third of overall suicides around the globe [3, 4], and PQ has a higher case fatality ratio than other commonly ingested pesticides [5]. Forty milliliters of a 24% PQ solution are enough to cause multiple organ failure and mortality within days [6]. Although the introduction of formulation changes (a blue color, a stenching agent, and an emetic) to the liquid concentrate, mortality after intentional ingestion remains high [7].

PQ poisoning is generally associated with suicide, occupational, and accidental exposure [8]. Homicides induced by PQ, in contrast, are rare and often associated with ingestion or direct skin contact [9–11]. The primary injury caused by PQ to mammalian systems occurs in the lung, and other major organ systems are also involved, but to a lesser extent [12]. Regardless of its route of administration, PQ is rapidly distributed in most tissues [12]. However, reports on PQ concentrations in common body fluids and major organ are rare.

Here, we report a rare case of homicidal PQ poisoning by oral administration. A 37-year-old man intoxicated from drinking alcohol drank white wine mixed with PQ prepared by his wife, but was not aware of it, blamed his symptoms on the intoxication and eventually died 22 days later. The clinical and autopsy results of the case are presented. We also present the PQ concentrations of 16 postmortem specimens (body fluids and tissues) in order to accumulate data on PQ distribution.

1.1 | Case history

A 37-year-old man was admitted to a county hospital due to intermittent vomiting 3 days post drinking and chest tightness with numbness in both lower extremities over the last 7 h. Blood examination revealed acute renal failure and a high level of amylase: white blood cells amounting to 17.6 × 10^9/L (Reference range: 3.5–9.5 × 10^9/L), neutrophils amounting to 15.7 × 10^9/L (Reference range: 1.8–6.3 × 10^9/L), creatinine amounting to 696 μmol/L (Reference range: 53–123 μmol/L), urea amounting to 17.2 mmol/L (Reference range: 2.8–8.3 mmol/L), and amylase amounting to 738.3 U/L (Reference range: 0–128 U/L). Color ultrasound revealed that the volume of both kidneys increased, Endoscopy revealed esophagitis and erosive gastritis, and his chest computed tomography (CT) scan showed bilateral infiltration distributing with partial interstitial changes of the lung and multiple gas accumulations in bilateral neck and mediastinum. The chief medical priorities were symptomatic treatment, venous nutrition support, and protection of gastric mucosa.

However, the patient did not improve and developed a fever (38.8 °C) 2 days after admission. His relatives transferred him to a municipal hospital for further treatment. Blood biochemistry analysis 1 day after admission showed acute liver insufficiency: Alanine transaminase amounting to 333 U/L (Reference range: 0–40 U/L), Aspartate aminotransferase amounting to 307 U/L (Reference range: 0–40 U/L), blood ammonia amounting to 72.20 μmol/L (Reference range: 16–60 μmol/L). In consideration of deficient liver and kidney functions of the patient, dialysis therapy was recommended. Due to the financial pressure, the patient had to be transferred back to the former county hospital. Through dialysis treatment, hormone shock, and other treatments, the patient's liver and kidney function improved, but the pulmonary interstitial lesions aggravated, and the blood oxygen saturation became increasingly difficult to maintain. Unfortunately, the patient developed multiple organ failure and eventually died of respiratory failure 19 days after the first admission. Postmortem specimens of cardiac blood, peripheral blood, cerebrospinal fluid, vitreous humor, urine, bile, pericardial effusion, pleural effusion, lung, liver, kidney, heart, spleen, pancreas, brain, and muscle were collected. All biological specimens were subsequently sent to the laboratory for PQ quantitative analysis (High-performance liquid chromatography-tandem mass spectrometry; HPLC-MS/MS). The timeline of this case was shown in Figure 1.

1.2 | Toxicological Analysis

1.2.1 | Chemicals and reagents

Paraquat (99.0%) was purchased from Dr. Ehrenstorfer GmbH (Augsburg, Germany). Formic acid and ammonium acetate (HPLC grade) were bought from CNW Technologies (Dusseldorf, Germany). Acetonitrile (HPLC grade) was obtained from Merck (Darmstadt, Germany). It was prepared and diluted with ultrapure water (18.2 MΩ, 25°C) from Milli-q water purification system (Merck Millipore, MA, USA) for all aqueous solutions.

1.2.2 | Sample preparation

The lung, liver, kidney, heart, spleen, pancreas, brain, and muscle specimens were first homogenized in deionized water. A 500-μL sample was placed in a 2-mL Eppendorf tube, mixed with 1 mL of acetonitrile, and then centrifuged at 13000 rpm for 5 min, and the supernatant was discarded. One milliter of 0.1% formic acid aqueous solution was added to the precipitate, and then centrifuged at 13000 rpm for 5 min, and the supernatant was sampled.

1.2.3 | HPLC-MS/MS conditions

The samples were analyzed by HPLC-MS/MS (LC-20A, Shimadzu Co., Kyoto, Japan) with a Zorbax XDB C-18 column (75 mm × 2.1 mm, 3.5 μm, Agilent, USA). The mobile phase consisted of two solvents: mobile phase A was acetonitrile and mobile

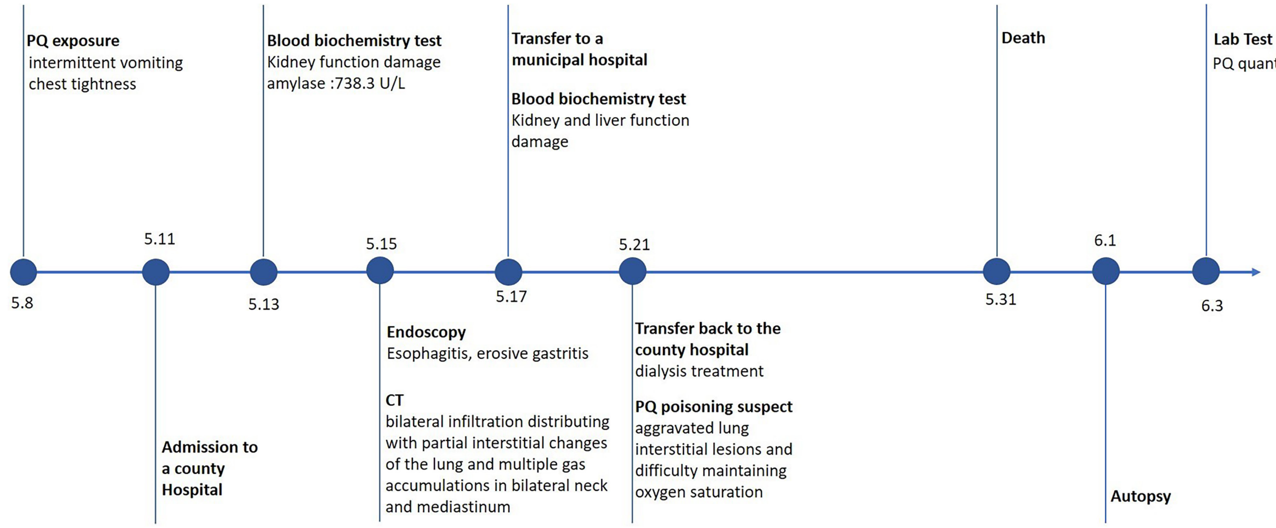

FIGURE 1 The timeline of case history [Color figure can be viewed at wileyonlinelibrary.com]

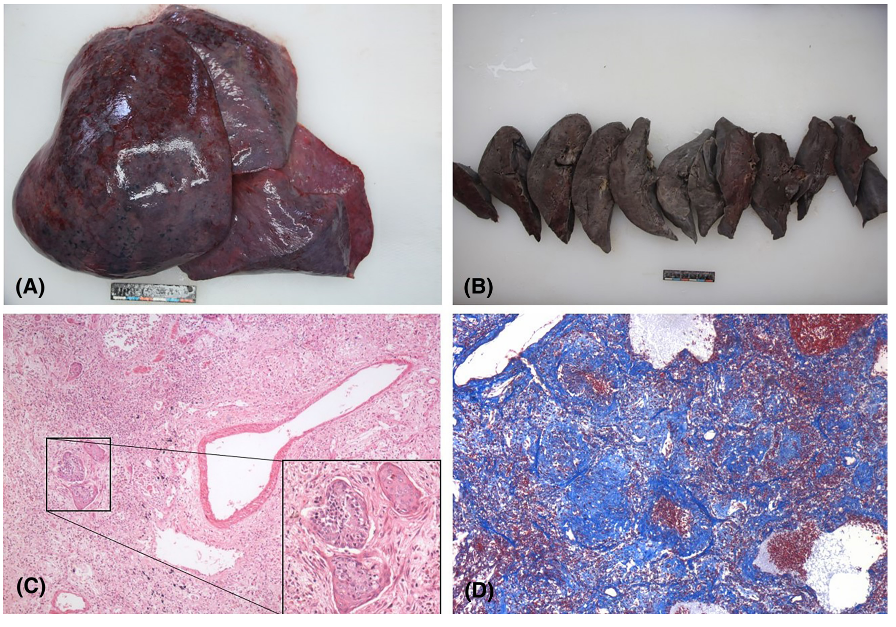

FIGURE 2 The gross and micrographs of lungs. (A) bleeding on the surface of the lung; (B) the heavy and firm lung parenchyma; (C) severe pulmonary interstitial fibrosis (HE, 40×), lung squamous metaplasia (HE, 100×); (D) massive fibrous connective tissue proliferation (Masson, 40×) [Color figure can be viewed at wileyonlinelibrary.com]

phase B was 20 mM of ammonium acetate aqueous solution with 0.1% formic acid. The flow rate was 0.40 mL/min and the column temperature was 40 °C.

The mass spectrometry was performed in positive ion multiple reaction monitoring (MRM) mode. The conditions used for mass spectrometric detection were as follows: IonSpray voltage, 5.5 kV; source temperature, 500 °C. The injection volume was 5 μL. Two transition ions of paraquat were monitored, the one with higher abundance (m/z 186.3–158.9) for quantification and the other (m/z 186.3–144.8) for confirmation.

1.2.4 | Paraquat quantification

A HPLC-MS/MS analysis combined with external standard working curve method was applied to PQ quantitation. The assay showed a linear response for PQ (20–500 ng/mL, r2 = 0.9986). Limits of detection/quantification of the method were 5 and 20 ng/mL.

2 | RESULTS

2.1 | Autopsy findings

The autopsy was performed 2 days post-death. There was no sign of trauma or injury except for several injection points. The fixed left lung weighed 590 g and the right lung weighed 710 g. Diffuse bleeding spots were found on the lung surface, and the lung parenchyma was heavy and firm with signs of severe congestion (Figure 2A, B). The most prominent histological finding was severe pulmonary interstitial fibrosis accompanied by capillary congestion, reactive pneumocyte changes, hyaline membranes, and squamous

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metaplasia corresponding to proliferative diffuse alveolar damage (Figure 2C). Masson trichrome stains showed destruction of normal lung structure and massive fibrous connective tissue proliferation (Figure 2D). The pulmonary interstitium and the alveolar spaces showed multiple focal hemorrhage and infiltration of inflammatory cells dominated by lymphocytes. Massive neutrophils were found in the hilar lymph nodes. The left pleural effusion was 400 ml, and the right pleural effusion was 300 ml.

The brain was found to be edematous and histologically the brain showed neuronal necrosis, increased proportion of inflammatory cells in the vessels with predominantly neutrophils, a small amount of lymphocyte infiltration around the vessel wall, widening of the VICHOW-ROBIN spaces, and sparing of the surrounding brain tissue. Sparse brain tissue with myelin loss was observed in the subventricular ventricles (Figure 3A). The tongue, the epiglottis, the esophagus, and the larynx were swollen and the mucosa of the tongue and the epiglottis is pale in appearance. Fifty microliter dark brown stomach contents were found. Microscopically, diffuse neutrophil infiltrated the musculature of the esophagus with focal muscle necrosis and some glands of the esophagus (Figure 3B) and the larynx showed squamous metaplasia. Scattered infiltration of lymphocytes was observed in the submucosa of esophagus, larynx, stomach, and intestine (Figure 3C, D).

A large number of inflammatory cells mainly composed of neutrophils were observed in the papillary interstitium and some papillary muscle fibers were necrotic and dissolved (Figure 3E). Surface and section of the enlarged kidney (left:440 g; right: 380 g) were congested. Histologically, kidney showed acute tubular damage and multiple focal hemorrhage with infiltration of scattered inflammatory cells dominated by lymphocytes. Granular cast, red cell cast, and hyaline cast were found in the lumen of renal tubules. Focal hemorrhage was observed in the myocardial interstitium, scattered inflammatory cells, mainly neutrophils, were found in the papillary interstitium and some of the myocardium was necrotic and lysed. The liver was firm and grayish-white in color. Histologically, the hepatocytes showed centrilobular necrosis, focal vacuolar degeneration, and cholestasis. Lymphocytic at portal areas increased and interlobular biliary epithelial cells appeared edematous and degenerated. The pancreatic stroma was scattered with few lymphocytes (Figure 3F). The neutrophils in the perifollicular and marginal zones of the spleen were increased.

2.2 | Analytical findings

The concentrations of paraquat in 16 specimens are presented in Table 1. All tissues tested positive for PQ. The concentration of PQ in postmortem specimens from high to low is kidney>urine > liver>lung> muscle>bile> heart>brain = pancreas>spleen>

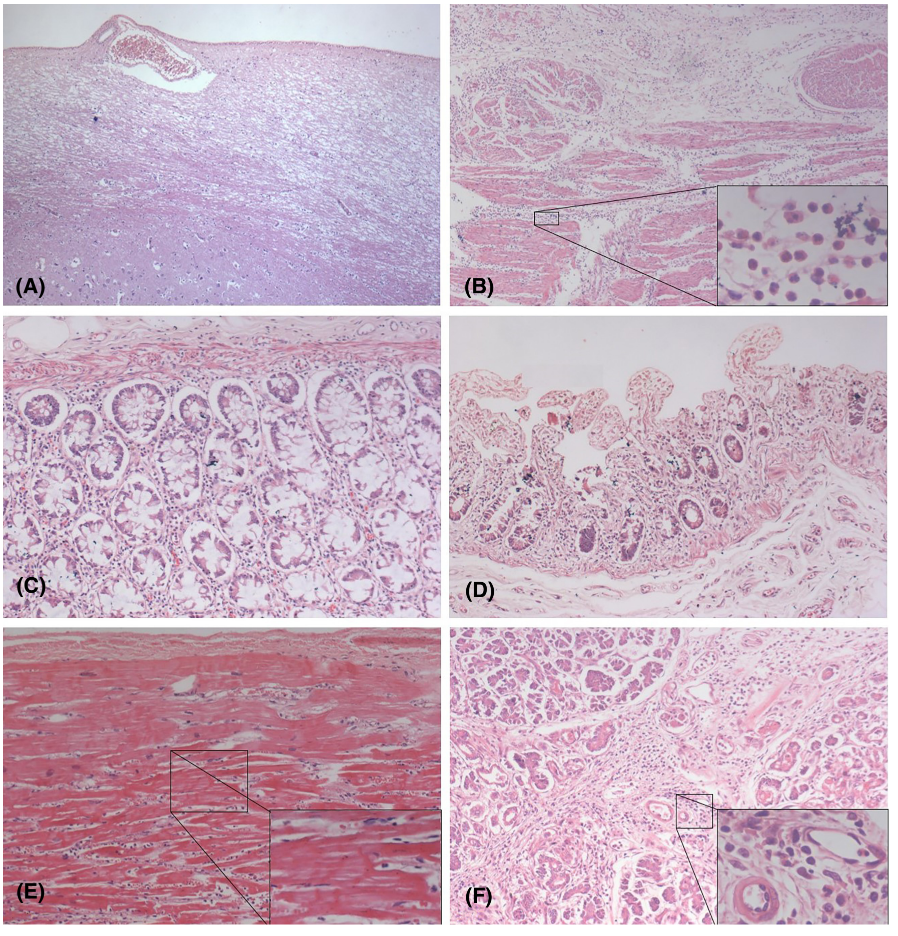

FIGURE 3 The micrographs of brain, esophagus, papillary muscle, pancreas, small intestine, large intestine. (A) sparse brain tissue with myelin loss in the subventricular ventricles (HE, 40×); (B) inflammatory cells exudation in esophageal and necrotic muscle fibers (HE, 40×, 400×); (C) and (D) inflammatory cells exudation in large intestine (HE, 100×)and small intestine (HE, 40×); (E) inflammatory cells exudation in the papillary interstitium and necrotic papillary muscle fibers (HE, 100×, 200×); (F) inflammatory cells exudation in pancreas interstitium (HE, 100×, 400×) [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 PQ concentrations in postmortem specimens in this case and in the literature

	Vandenbogaerde [13]	Fernandez [14]	Dinis-Oliveira [15]					Moreira [16]		CHEN [11]	The present case
kidney	10.50	8.61	5.662	0.575	1.044	1.145	0.998	3.68	5.23	0.24	1.31
urine		16.5	13.539	0.078	0.592	0.177	0.500	3.94	11.04		0.91
liver	1.32	4.36	4.355	0.340	0.973	0.879	0.100	1.78	6.33	0.20	0.62
lung	3.23	6.24	11.856	0.660	5.095	2.231	0.500	6.24	8.23	0.49	0.39
muscle											0.35
bile											0.32
heart	3.00		0.023	0.001		0.007	not detected				0.28
brain										0.32	0.22
pancreas											0.22
spleen											0.18
cardiac blood			9.500	not detected	0.290	0.090	not detected			0.11	0.15
cerebrospinal fluid											0.14
pericardial effusion											0.12
pleural effusion											0.09
peripheral blood											0.08
vitreous humor											0.06
plasma		0.10									
whole blood								0.97	1.82		
duodenal wall			3.100	0.260	2.320	0.168	2.830				
gastric wall			1.234	0.190	0.200	0.094	0.160			<0.05	
diaphragm			0.305	0.016	0.130	0.131	0.010				
Survival time after ingestion	6d	16d	9h	4d	2d	4d	6d	unknown	unknown	24d	22d

cardiac blood>cerebrospinal fluid>pericardial effusion>pleural effusion>peripheral blood>vitreous humor.

3 | DISCUSSION

Extensive pulmonary fibrosis with aseptic inflammation and renal insufficiency and the detection of PQ in multiple postmortem specimens in this case was consistent with the diagnosis of PQ poisoning [1]. PQ poisoning is the cause of death and respiratory failure resulted from lung injury is the mechanism of death. Since the patient survived for more than 20 days after PQ poisoning, the lung mainly showed hyperplastic changes and even squamous metaplasia, indicating severe lung injury. PQ concentrations in postmortem specimens in this case and in the literature [11, 14–17] are presented in Table 1, indicating that PQ concentrations accumulated in different tissues and body fluids in humans are in different levels. In the presented case, all tissues tested positive for PQ and the main organs in which PQ accumulated were kidney, liver and lung. In spite of repeated hemodialysis, PQ was re-released and redistributed from the accumulated PQ in various tissues of the deceased, making it still detectable in postmortem tissues.

As the main excretory organ of PQ, the kidney is the organ with the highest concentration of PQ, followed by the urine. In humans, over 90% of the absorbed dose of PQ is excreted unchanged via urine within 12 to 24 h consistent with the high levels of PQ detected in the urine [18]. According to previously reported studies, PQ is primarily eliminated by tubular filtration and active tubular secretion in human body [19], tubular reabsorption being minimal [20]. Ingestion of large doses of PQ results in tubular necrosis with a rapid decrease in GFR and tubular secretion and causes a prolonged elimination half-life [19, 21]. In contrast, some studies showed the highest concentration of PQ in lung [11, 16, 17], which might be related to the rapid fatal intoxication and relatively mild degree of kidney injury. The main damage caused by PQ to mammalian systems occur in the lung, where PQ is accumulated through an active transport process in the Clara cells and alveolar type I and type II epithelial cells [12]. When Clara cells and alveolar type I and II epithelial cells were massively necrotic and lung fibrosis progressed, the efficiency of the active transport process decreased and therefore PQ concentration in the lung reduced. The detection results, clinical biochemical indicators, and pathological results of liver damage also indicate that the liver is one of the main organs involved in the accumulation of PQ poisoning. Despite the decrease of PQ in the cardiac blood and the peripheral blood in both literature and this case, organs such as liver, lungs, and spleen still retain PQ for a long time, which may be due to the continuous absorption of PQ from the intestine or the existence of PQ hepatic-intestinal circulation consistent with the presence of PQ in the bile. The presence of PQ in the stomach and duodenal wall reported in the literature is in accordance with these hypotheses [16].

An in vivo PET study in rhesus macaque revealed that PQ was excluded by the blood brain barrier (BBB) [22], but PQ was detected in the cerebrospinal fluid and the brain in this case, which might indicate that continuous PQ exposure or high dose PQ could damage the blood-brain barrier and then allow PQ to enter the brain. Meanwhile, the in vivo PET study also showed that the remaining cerebral PQ was mostly confined to the pineal gland and the lateral ventricles after administration, both of which lack a BBB [22]. The autopsy studies of Grant [23] and Hughes [24] also reported periventricular damage induced by PQ. Thus, PQ may damage the brain through the weaker parts of the blood-brain barrier (such as the ventricles), as evidenced by the detection of higher concentrations of PQ in cerebrospinal fluid. The detection of PQ in the heart and the pathological changes of interstitial myocardial hemorrhage and papillary myocarditis confirmed the myocardial toxicity of PQ. Povoa et al. [25] reported that cardiac toxicity due to PQ is frequent (40%). Conversely, the PQ concentrations of the heart were low in the five cases of PQ poisoning reported by Dinis-Oliveira, which may be associated with lower doses or rapid lethality [16]. Esophageal smooth muscle necrosis with a large number of neutrophils infiltration might cause a small leak in the esophagus, related to the corrosive and the oxidative stress effects of PQ and vomiting. Multiple gas accumulation in the bilateral neck and mediastinum on CT further demonstrated the presence of a small leak, which also suggest a potential esophageal perforation. Ackrill et al. [26] reported 2 cases of esophageal perforation due to PQ that no definite perforation was found, but a small leak in the esophagus resulted in mediastinitis and death. Esophageal treatment of PQ poisoning should be paid enough attention.

PQ poisoning often led to an increase in serum amylase levels and the severity of the intoxication was found to be related to elevated serum amylase levels, and hyperamylasemia has been used as an early predictor of mortality in patients with acute paraquat poisoning [27]. Nonetheless, the serum amylase was significantly elevated in this case, but significant abdominal pain was not observed, and repeated abdominal ultrasonography and CT from admission to death did not reveal pancreatic lesions and histologically only mild inflammatory cell infiltration was found in the pancreatic stroma without obvious pancreatic necrosis. In fact, serum amylase consisted of P-type amylase and S-type amylase, and the pancreas secretes P-type amylase in vivo, while extra-pancreatic organs secrete S-type amylase [28]. The elevation of serum amylase may be caused not only by acute pancreatitis but also by acute renal failure, mumps, esophagitis, vomiting, and other factors [29]. In this case, gastrointestinal damage and inflammation due to paraquat toxicity can increase S-type amylase, and the concentration of the total amylase possibly amplified by impaired renal clearance. Postmortem serum amylase and isoenzyme assays were performed, and the total amylase concentration in the postmortem serum was 580 U/L and the pancreatic amylase concentration was 68 U/L. Michiue's study [30] showed that the cut-off values for the postmortem left and right cardiac serum AMY to discriminating intoxication and other groups were 550 U/L and 450 U/L, respectively, and serum S-type amylase levels showed a tendency to increase more significantly in the toxic group, consistent with the results of this case. Elevated serum amylase may be explained as follows: 1. mild pancreatic damage; 2. impaired renal clearance; 3. gastrointestinal damage.

The incidents of PQ poisoning after drinking are not uncommon, and patients who took PQ after drinking have been reported to be successfully saved [31], which gave rise to the idea that ethanol might reduce the damaging effects of PQ. However, in this case, the severe renal failure and inflammatory response of the deceased 3 days post ingestion of PQ and the subsequent irreversible pulmonary fibrosis do not support a protective effect of ethanol against PQ poisoning. Meanwhile, the detection of PQ in various postmortem tissues and body fluids showed that ethanol did not inhibit the distribution and accumulation of PQ. In fact, it has been reported that combined ingestion of both ethanol (2 or 3.8 g/kg) and PQ (200 mg/kg) resulted in increased PQ levels in the blood compared to animals ingesting just PQ [32], demonstrating the interaction between alcohol and PQ. Further animal study suggested that ethanol might potentiate PQ toxicity by increasing intestinal absorption and tissue distribution (33). In practice, ethanol can lead to congestion and erosion of the mucous membrane of the gastrointestinal tract, which may weaken the barrier effect of the gastrointestinal tract and aggravate the corrosive effect of PQ on the gastrointestinal tract and other organs. The effect of alcohol consumption on the prognosis and mortality of PQ poisoning and the specific mechanism still need further study.

Occult PQ poisoning is initially difficult to diagnose, especially when the exposure history is unclear, because the symptoms are mild and non-specific. In this case, the diagnosis of PQ poisoning was not recognized until pulmonary fibrosis developed, but it was too late to prevent the disease from worsening, even with plasmapheresis. In general, for PQ poisoning, oral administration will cause upper gastrointestinal irritation and more or less damage to the tongue and oral mucosa, while skin contact will cause skin damage, and at the same time, blood examination will show elevated inflammatory indicators, followed by kidney, liver, and lung damage. Thus, nowadays, although PQ has been banned in many countries, there are still small quantities of it in stock and illegally manufactured, which still needs to be carefully identified in clinical diagnosis and forensic practice.

REFERENCES

1. Dinis-Oliveira RJ, Duarte JA, Sanchez-Navarro A, Remiao F, Bastos ML, Carvalho F. Paraquat poisonings: Mechanisms of lung toxicity, clinical features, and treatment. Crit Rev Toxicol. 2008;38(1):13–71. https://doi.org/10.1080/10408440701669959
2. Wilks MF, Fernando R, Ariyananda PL, Eddleston M, Berry DJ, Tomenson JA, et al. Improvement in survival after paraquat ingestion following introduction of a new formulation in Sri Lanka. PLoS Med. 2008;5(2):e49. https://doi.org/10.1371/journal.pmed.0050049
3. Gunnell D, Eddleston M, Phillips MR, Konradsen F. The global distribution of fatal pesticide self-poisoning: systematic review. BMC Public Health. 2007;7(1):357. https://doi.org/10.1186/1471-2458-7-357
4. Bertolote JM, Fleischmann A, Butchart A, Besbelli N. Suicide, suicide attempts and pesticides: A major hidden public health problem. Bull World Health Organ. 2006;84(4):260. https://doi.org/10.2471/blt.06.030668
5. van der Hoek W, Konradsen F. Analysis of 8000 hospital admissions for acute poisoning in a rural area of Sri Lanka. Clin Toxicol (Phila). 2006;44(3):225–231. https://doi.org/10.1080/15563650600584246
6. Yang CJ, Lin JL, Lin-Tan DT, Weng CH, Hsu CW, Lee SY, et al. Spectrum of toxic hepatitis following intentional paraquat ingestion: Analysis of 187 cases. Liver Int. 2012;32(9):1400–1406. https://doi.org/10.1111/j.1478-3231.2012.02829.x
7. Meredith T, Vale JA. Treatment of paraquat poisoning: gastrointestinal decontamination. In: Bismuth C, Hall AH, editors. Paraquat poisoning: Mechanisms, prevention, treatment. New York, NY: Marcel Dekker; 1995. p. 297–314.
8. Yin Y, Guo X, Zhang SL, Sun CY. Analysis of paraquat intoxication epidemic (2002–2011) within China. Biomed Environ Sci. 2013;26(6):509–512. https://doi.org/10.3967/0895-3988.2013.06.014
9. Stephens BG, Moormeister SK. Homicidal poisoning by paraquat. Am J Forensic Med Pathol. 1997;18(1):33–39. https://doi.org/10.1097/00000433-199703000-00006
10. Daisley H, Simmons V. Homicide by paraquat poisoning. Med Sci Law. 1999;39(3):266–269. https://doi.org/10.1177/002580249903900314
11. Chen F, Ye Y, Jin B, Yi B, Wei Q, Liao L. Homicidal paraquat poisoning. J Forensic Sci. 2019;64(3):941–945. https://doi.org/10.1111/1556-4029.13945
12. Suntres ZE. Role of antioxidants in paraquat toxicity. Toxicology. 2002;180(1):65–77.
13. Vandenbogaerde J, Schelstraete J, Colardyn F, Heyndrickx A. Paraquat poisoning. Forensic Sci Int. 1984;26(2):103–114. https://doi.org/10.1016/0379-0738(84)90066-5
14. Fernandez P, Bermejo AM, Lopez-Rivadulla M, Cruz A, Rodriguez E, Otero A. A fatal case of parenteral paraquat poisoning. Forensic Sci Int. 1991;49(2):215–224. https://doi.org/10.1016/0379-0738(91)90082-t
15. Dinis-Oliveira RJ, de Pinho PG, Santos L, Teixeira H, Magalhaes T, Santos A, et al. Postmortem analyses unveil the poor efficacy of decontamination, anti-inflammatory and immunosuppressive therapies in paraquat human intoxications. PLoS One. 2009;4(9):e7149. https://doi.org/10.1371/journal.pone.0007149
16. Moreira PN, de Pinho PG, Baltazar MT, Bastos ML, Carvalho F, Dinis-Oliveira RJ. Quantification of paraquat in postmortem samples by gas chromatography-ion trap mass spectrometry and review of the literature. Biomed Chromatogr. 2012;26(3):338–349. https://doi.org/10.1002/bmc.1663
17. Houze P, Baud FJ, Mouy R, Bismuth C, Bourdon R, Scherrmann JM. Toxicokinetics of paraquat in humans. Hum Exp Toxicol. 1990;9(1):5–12. https://doi.org/10.1177/096032719000900103
18. Bismuth C, Baud FJ, Garnier R, Muszinski J, Houze P. Paraquat poisoning: Biological presentation. J Toxicol Clin Exp. 1988;8(3):211–218.
19. Beebeejaun AR, Beevers G, Rogers WN. Paraquat poisoning-prolonged excretion. Clin Toxicol. 1971;4(3):397–407. https://doi.org/10.3109/15563657108990492
20. Bismuth C, Scherrmann JM, Garnier R, Baud FJ, Pontal PG. Elimination of paraquat. Hum Toxicol. 1987;6(1):63–67. https://doi.org/10.1177/096032718700600110
21. Bartlett RM, Holden JE, Nickles RJ, Murali D, Barbee DL, Barnhart TE, et al. Paraquat is excluded by the blood brain barrier in rhesus macaque: An in vivo pet study. Brain Res. 2009;1259:74–79. https://doi.org/10.1016/j.brainres.2008.12.033
22. Grant H, Lantos PL, Parkinson C. Cerebral damage in paraquat poisoning. Histopathology. 1980;4(2):185–195. https://doi.org/10.1111/j.1365-2559.1980.tb02911.x
23. Hughes JT. Brain damage due to paraquat poisoning: A fatal case with neuropathological examination of the brain. Neurotoxicology. 1988;9(2):243–248.

JOURNAL OF FORENSIC SCIENCES AAFS

24. Povoa R, Maciel FM, Orlando JM, Povoa EF, Honsi E, Kasinski N, et al. A cardiac lesion secondary to paraquat. Arq Bras Cardiol. 1992;59(2):95–98.
25. Ackrill P, Hasleton PS, Ralston AJ. Oesophageal perforation due to paraquat. Br Med J. 1978;1(6122):1252–1253. https://doi.org/10.1136/bmj.1.6122.1252
26. Li Y, Wang M, Gao Y, Yang W, Xu Q, Eddleston M, et al. Abnormal pancreatic enzymes and their prognostic role after acute paraquat poisoning. Sci Rep. 2015;5(1):17299. https://doi.org/10.1038/srep17299
27. Pieper-Bigelow C, Strocchi A, Levitt MD. Where does serum amylase come from and where does it go? Gastroenterol Clin North Am. 1990;19(4):793–810.
28. Huang C, Bai L, Xue X, Peng L, Jiang J, Zhang X. Hyperamylasemia as an early predictor of mortality in patients with acute paraquat poisoning. J Int Med Res. 2020;48(3):300060520910037. https://doi.org/10.1177/0300060520910037
29. Michiue T, Ishikawa T, Kawamoto O, Chen JH, Wang Q, Zhu BL, et al. Postmortem serum levels of amylase and gamma glutamyl transferase (GGT) as markers of systemic tissue damage in forensic autopsy. Leg Med (Tokyo). 2013;15(2):79–84. https://doi.org/10.1016/j.legalmed.2012.09.003
30. Ragoucy-Sengler C, Pileire B, Daijardin JB. Survival from severe paraquat intoxication in heavy drinkers. Lancet. 1991;338(8780):1461. https://doi.org/10.1016/0140-6736(91)92762-q
31. Kuo TL, Nanikawa R. Effect of ethanol on acute paraquat toxicity in rabbits. Nihon Hoigaku Zasshi. 1990;44(1):12–17.
32. Yu HY, Lai YR, Kuo TL, Shen YZ. Effects of ethanol on pharmacokinetics and intestinal absorption of paraquat in animals. J Toxicol Sci. 1994;19(2):67–75. https://doi.org/10.2131/jts.19.2_67

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